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Towards Agentic Urban Digital Twins (AUDiTs): advancing new urban science through Human-AI co-learning agents

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Abstract

While digital twins provide predictive and planning capabilities, they often underrepresent social complexity, ethical considerations, and stakeholder participation. Concurrently, advances in artificial intelligence offer new opportunities for real-time sensing, adaptive learning, and decision support; yet current applications remain narrowly focused on optimization or monitoring. This falls short of the broader mission of New Urban Science: to Explain, Discover, Understand, and Generalize (EDUG) complex socio-technical systems. This vision paper introduces Agentic Urban Digital Twins (AUDiTs), a research agenda for integrating large language model (LLM) and multimodal agents into digital twin environments to foster human and AI co-learning, context-aware reasoning, participatory scenario design, and ethical deliberation. We identify critical challenges such as bias and fairness in foundation models, limitations in data-scarce contexts, computational sustainability, and institutional alignment, and we outline pathways for addressing them. AUDiTs extend existing digital twin paradigms by shifting emphasis from purely technical prediction toward collaborative, explainable, and value-sensitive urban intelligence, positioning digital twins as platforms for responsible and adaptive human and AI partnership in shaping just and resilient urban futures.

Keywords Agentic Urban Digital Twins, New Urban Science, Human-Centric AI, Human Dynamics

1 Introduction

Artificial intelligence (AI) is reshaping urban science, by expanding what can be observed, modeled, and managed in cities. From processing large geospatial datasets to simulating complex spatial and social dynamics, modern AI technologies such as deep neural networks and large language and multimodal transformer models enable analytical capacities beyond traditional approaches (He & Chen, 2024). Earlier urban models relied on static assumptions and manual rules. AI-driven systems support dynamic, adaptive, and cross-domain modeling (Liu et al., 2025). These advances raise a key question: How can AI be used not only for automation but also to advance the mission of urban science to Explain, Discover, Understand, and Generalize (EDUG) human-centered urban systems (Ye et al., 2025)?

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Addressing this question requires rethinking the role of AI in urban research and practice. New Urban Science is not a rebranding of urban analytics but a paradigm that combines computational intelligence with ethical, participatory, and system-oriented design. Instead of separating technical innovation from societal context, it emphasizes human and AI co-learning, value-sensitive system design, and urban systems thinking. Cities are understood not as mechanistic infrastructures but as adaptive socio-technical ecosystems that connect people, place, and policy (Monstadt & Coutard, 2019). This framing builds on Batty's *New Science of Cities* (Batty, 2013), which sees cities as evolving systems of networks and flows shaped by collective action, while also recognizing a longer intellectual lineage from early visions of synthetic urban agents in science and fiction to current research in multi-agent urban modeling (An et al., 2021; Ligmann-Zielinska et al., 2020). Building on this foundation, New Urban Science advances the EDUG mission through AI-enabled, ethically grounded, and participatory approaches that align societal values with technical innovation.

One frontier where this synthesis becomes tangible is the evolution of urban digital twins (Ye et al., 2023). Existing digital twins already support monitoring, forecasting, and planning, but they are often application-specific, infrastructure-centric, and limited in their capacity for reasoning, adaptation, and stakeholder interaction (Willcox et al., 2023). To move beyond these constraints, we introduce Agentic Urban Digital Twins (AUDiTs), a research agenda that explores how AI-enabled, agentic frameworks can augment digital twins with generative modeling, context-aware reasoning, and interactive, participatory interfaces. Rather than implying that large language models are intrinsically human-centric, AUDiTs situate such models within human-in-the-loop and institutionally guided processes, enabling cities to evolve as platforms for responsible and collaborative human–AI partnership.

Recent work underscores the need for standardized interoperability frameworks that enable collaboration among heterogeneous AI agents and model types. As Ehtesham et al. (2025) observe, large language model powered autonomous agents require robust, standardized protocols to integrate tools, share contextual data, and coordinate tasks across distributed systems. Among the emerging solutions, the Model Context Protocol (MCP) provides a secure JSON-RPC interface for tool access and structured data exchange, while the Agent-to-Agent (A2A) protocol supports peer-to-peer task delegation through capability-based credentials for scalable, cross-domain collaboration. Complementing these standards, AutoGen provides a high-level orchestration framework for constructing multi-agent systems in

which customizable and conversable agents that combine LLMs, human inputs, and external tools can collaborate to accomplish complex tasks (Wu et al., 2024). Within this broader Hybrid Multi-Model Architecture, diverse AI paradigms such as LLMs, LVMs, diffusion models, RL/IL, causal inference, and symbolic reasoning contribute complementary strengths in perception, reasoning, and control (Bommasani, 2021). Together, these advances form the technical foundation for AUDiTs, enabling interoperable, ethically aligned, and verifiable human–AI co-learning ecosystems that integrate multimodal urban data streams and decision processes across domains.

Figure 1 illustrates the conceptual framework of AUDiTs. AUDiTs differ from conventional digital twins in three fundamental dimensions: 1) Embedded agency, allowing for multi-agent simulation, strategic adaptation, and behavioral modeling; 2) Co-learning workflows, enabling mutual adaptation between AI systems and human stakeholders; 3) Value alignment, supporting transparency, contestability, and ethical reasoning in decision-making processes.

2 Definitions

We define AUDiTs as next-generation digital representations of urban systems, augmented by intelligent agents powered by LLMs and autonomous geospatial intelligence. Unlike conventional digital twins, which already support monitoring, forecasting, and planning, AUDiTs extend their scope by embedding reasoning capabilities through LLM-augmented geographic information systems (GIS) and agentic orchestration. They are explicitly designed to advance the EDUG mission of urban science—Explaining, Discovering, Understanding, and Generalizing human-centered urban phenomena. The proposed framework emphasizes not only data acquisition, spatial pattern detection, and scenario simulation, but also interpretability, ethical alignment, and participatory interaction at scales and levels of abstraction not previously achievable.

A distinctive strength of AUDiTs lies in their combined capacity for data driven and knowledge driven reasoning. In data rich settings such as real time traffic telemetry, deep learning models can identify patterns and forecast system behavior. In contrast, in data sparse or ethically sensitive domains such as eviction records or environmental health disparities, symbolic AI and structured domain knowledge are essential for inference, transparency, and fairness. At the same time, current limitations of large language models must be acknowledged, including bias, reliability concerns, and difficulty generalizing from incomplete or uneven urban data. For this reason, AUDiTs should be viewed not as a turnkey system but as a research agenda for integrating human and

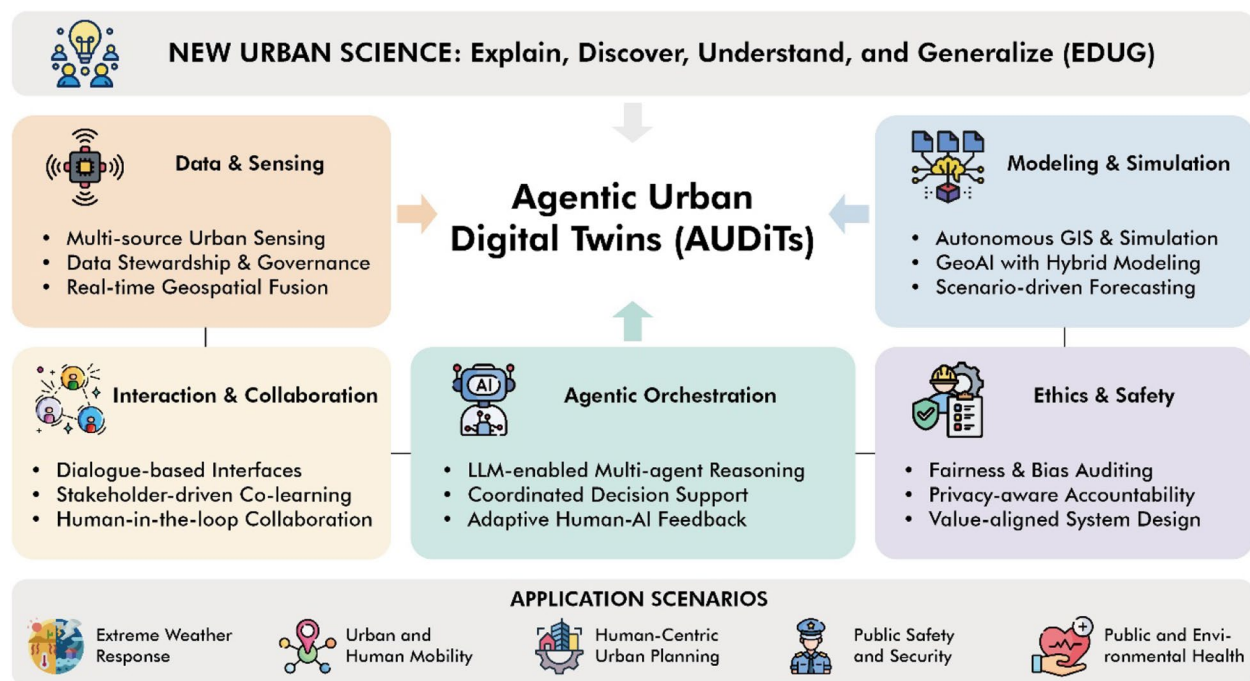


Fig. 1 Conceptual framework of Agentic Urban Digital Twins

AI collaboration, ethical safeguards, and cross domain reasoning within digital twin platforms. In this sense, AUDiTs reposition the role of AI in cities as an agentic partner that works with human stakeholders to reason, learn, and support the design of more just and resilient urban futures.

3 Key components

The AUDiT systems are structured into five interdependent components: Data & Sensing, Modeling & Simulation, Agentic Orchestration, Interaction & Collaboration, and Ethics & Safety. Together, these components combine human knowledge with machine intelligence to create a scalable, transparent, and ethically grounded decision-support system for cities. Table 1 presents a compact, end-to-end pipeline view of these components, summarizing their inputs, assumptions, outputs, and

limitations and indicating their maturity using status markers included in the table: I (implemented), E (emerging), and A (aspirational). The subsections that follow provide detailed explanations of each component, building on the high-level pipeline in Table 1 to clarify their specific roles within the integrated AUDiT framework.

3.1 Data & sensing

AUDiTs operate within complex, heterogeneous, and sensitive data environments that encompass multiple spatial and temporal scales. Effective implementation depends on integrating diverse sources such as IoT sensor feeds, satellite and aerial imagery, street-level LiDAR, participatory platforms, mobile traces, and administrative records. In addition to geospatial data, 3D modeling environments, agent-based simulation platforms, and physically based rendering systems must

Table 1 End-to-end AUDiT pipeline with key inputs, assumptions, outputs, and limitations across the five core components

Component	Status	Inputs	Outputs	Limits
Data & Sensing	I/E	Multimodal urban data streams	Unified geospatial dataset	Data gaps; privacy
Modeling & Simulation	E	Curated data; domain simulators	Predictive scenarios; visualizations	High compute; scaling
Agentic Orchestration	A	AI agents + LLM mediation	Coordinated reasoning, scenario planning	Reliability; fairness
Interaction & Collaboration	I/E	Dashboards; interfaces	Participatory decision spaces	Trust; access
Ethics & Safety	A	Audit tools; governance inputs	Bias checks; ethical interventions	Institutional variation

also be incorporated to support realistic, interactive, and multi-agent urban representations (Ye et al., 2023). A critical technical enabler is autonomous data fusion, which leverages multimodal AI models to harmonize structured and unstructured inputs into unified, dynamically updated geospatial frameworks. Ensuring responsible data use requires robust governance infrastructures that include standardized metadata protocols, transparent lineage tracking, and federated data-sharing agreements. Within this framework, the Data Curator Agent manages data acquisition, quality assurance, provenance, and bias detection, ensuring consistency across distributed data streams. Persistent challenges remain: data gaps in underrepresented communities, uneven interoperability across institutions, and the high computational demands of real-time ingestion continue to constrain scalability. In sensitive domains such as health, policing, and housing, AUDiTs must employ privacy-preserving techniques including synthetic data generation, differential privacy, and secure multiparty computation. While these methods provide partial safeguards, comprehensive solutions for protecting privacy while preserving analytical validity remain a critical area for ongoing research and innovation.

3.2 Modeling & simulation

At the core of AUDiTs is the autonomous GIS engine, a new class of AI-driven systems that can independently query geospatial data and model urban phenomena (Li et al., 2025). This engine extends conventional GIS toolkits by incorporating GeoAI models, such as machine learning algorithms and spatially explicit neural networks, to support both predictive analytics and causal inference. In practice, the engine can be coupled with cloud-based simulation platforms and connected through APIs to domain-specific simulators, including those for traffic flow, hydrology, and heat dispersion. This hybrid modeling environment enables researchers and practitioners to run virtual experiments, explore alternative “what-if” scenarios, and project how urban systems may change over time. Results can then be translated into decision-ready formats through dynamic visualization layers that support planning and stakeholder engagement. Despite these opportunities, important limitations remain. High-resolution simulations require substantial computing power and carefully calibrated parameters, which are not consistently available across all settings. In addition, scaling these models from city-level applications to regional or national systems continues to present significant challenges.

3.3 Agentic orchestration

At the core of AUDiTs is a multi-agent architecture that coordinates specialized AI agents through LLM mediation. This design allows agents to operate with partial autonomy while remaining connected through a shared knowledge base and adaptive feedback loops, much like how different information systems in a city must interoperate to support planning and governance. Implementation can build on emerging multi-agent frameworks such as LangChain or AutoGen, which are increasingly capable of modular orchestration, though their effectiveness at the scale and complexity of urban systems is not yet demonstrated (Wu et al., 2024). Within this architecture, the Data Curator Agent safeguards information quality by managing integrity, tracking uncertainty, and identifying bias across diverse urban datasets; the Scenario Planner Agent conducts simulations and policy experiments that mirror urban decision-making processes; the Dialogue Mediator Agent enables accessible, natural language interaction between stakeholders and the system; and the Ethics and Safety Agent monitors fairness and compliance to ensure that outcomes align with societal values. While these roles are conceptually robust, their technical maturity is uneven. Dialogue mediation with LLMs is already becoming feasible for tasks such as translating technical models into policy-relevant insights, whereas automated fairness auditing and policy-sensitive scenario planning remain under development and highlight the challenges of applying agentic systems responsibly in real urban contexts.

3.4 Interaction & collaboration

AUDIiTs are envisioned as collaborative infrastructures that embed human-in-the-loop processes into the orchestration of urban intelligence. Practical implementation involves the development of dashboards, conversational interfaces, and simulation sandboxes, all supported by cloud-based collaborative environments that allow multiple users to explore scenarios and adjust model assumptions together. These systems build on the legacy of participatory planning platforms such as Community-Viz or open-source civic engagement portals, but extend their capabilities through real-time AI reasoning and adaptive co-learning (Walker, 2017). The Dialogue Mediator Agent plays a key role in ensuring that interactions remain intelligible, inclusive, and reflexive. However, important limitations remain: cultivating trust in automated mediators, providing equitable access to digital participation, and balancing the contributions of experts and community members all continue to pose challenges for replicability and long-term legitimacy.

3.5 Ethics & safety

Ethical integrity and system safety are foundational pillars of the AUDiTs framework, ensuring that intelligent urban systems remain transparent, accountable, and socially embedded (Kitchin, 2021). In practice, ethical governance is operationalized through an Ethics & Safety Agent functioning as a hybrid mechanism combining automated auditing with participatory oversight. The Agent continuously monitors model behavior for bias, privacy risks, and disparate impact using fairness-aware AI pipelines and context-sensitive anomaly detection. When harm thresholds are approached, the Agent triggers human-in-the-loop interventions involving community review panels, municipal ethics boards, or civic data trusts, depending on local governance structures. This feedback process is designed to be adaptive to contested urban contexts, where ethical trade-offs between efficiency, equity, and privacy are negotiated through deliberation rather than automation. Technical enablers include model cards, decision logs, and explainability dashboards that document model lineage and decision criteria, as well as federated learning, secure multiparty computation, and policy ontologies encoding local legal and cultural norms. Because cities vary widely in institutional maturity and political culture, implementation must remain flexible—ranging from modular middleware in advanced digital twin infrastructures to community advisory mechanisms in resource-limited settings. Through this socio-technical integration, AUDiTs transform ethical safeguards from abstract principles into operational governance mechanisms, ensuring that digital twins embody value-sensitive, democratically accountable, and context-aware urban intelligence.

4 Application scenarios

AUDiTs are envisioned as co-learning ecosystems that bring together AI researchers, urban planners, community stakeholders, and scientists to advance ethical and evidence-based urban innovation. Emphasizing transparency, replicability, and participatory governance, they serve as living laboratories for responsible experimentation. While several foundational components—such as autonomous GIS engines, GeoAI modeling, and real-time sensing infrastructures—are already emerging in research and pilot projects, other capabilities, including multi-agent orchestration, ethics-aware mediation, and adaptive policy simulation, remain research frontiers. The illustrative use cases presented are hypothetical demonstrations intended to show how current developments could evolve into operational human–AI co-learning systems. Together, they outline pathways for connecting ongoing technical progress with long-term societal and

governance goals, advancing the transformative vision of AUDiTs within New Urban Science.

4.1 Extreme weather response

AUDiTs can function as climate resilience engines by dynamically integrating multi-hazard geospatial data such as floodplains, wildfire zones, and heat exposure together with real time meteorological inputs and population vulnerability indices. The Scenario Planner Agent, which remains an aspirational component, would simulate cascading impacts of extreme events under a range of adaptation strategies including green infrastructure, cooling corridors, and emergency shelter networks. In the near term, existing autonomous GIS and GeoAI frameworks already support limited coupling of climate and population data for scenario testing. During a heatwave, for example, a prototype system could merge mobility traces, land surface temperature, and demographic vulnerability to optimize the placement of cooling centers or prioritize energy delivery to at-risk communities. As adaptive learning agents mature, these capabilities could evolve from reactive crisis response toward anticipatory and sustained resilience planning.

4.2 Urban and human mobility

In mobility systems, many foundational elements of AUDiTs, such as real time sensing, traffic simulation, and AI routing algorithms, are already operational in smart city platforms. AUDiTs build on these capacities by adding agentic mediation and ethical oversight, which remain largely conceptual. The Autonomous GIS, an emerging capability, can already ingest vehicular sensors, GPS traces, and transit logs to support policy testing, including congestion pricing, smart signaling, or multi-modal integration. In the future, the Dialogue Mediator Agent could ensure that optimization remains transparent and equitable, allowing users to examine model assumptions, evaluate tradeoffs, and incorporate fairness constraints. This combination of existing and developing technologies enables co-designed mobility futures that improve efficiency while preserving inclusivity.

4.3 Human-centric urban planning

Digital twin platforms and participatory planning tools already support public engagement in urban design. AUDiTs extend these efforts by integrating GeoAI-driven simulation and human–AI co-learning, which remain in early prototype stages. Through scenario testing, planners and residents can evaluate how zoning, densification, or land-use changes affect affordability, walkability, and sustainability. Current visualization technologies enable partial versions of this process, while future multi-agent orchestration and adaptive reasoning could allow

stakeholders to collaboratively test design hypotheses in real time. In this way, AUDiTs democratize planning by merging quantitative simulation with qualitative insight, evolving static consultation into interactive co-design.

4.4 Public safety and security

The integration of spatial and temporal crime data, 911 call logs, and environmental features is already achievable with current predictive policing and hotspot analysis systems, although these tools often lack transparency and meaningful ethical safeguards. AUDiTs seek to reframe this practice through ethically constrained and participatory analytics. The envisioned Ethics and Safety Agent, which remains a conceptual design, would guide predictive insights toward preventive and restorative strategies rather than punitive responses. Early-stage implementations can already reveal correlations between infrastructure neglect and vulnerability, informing investments in public spaces, lighting, or social services. The longer-term goal is to incorporate fairness-aware AI pipelines that ensure public safety systems advance equity and spatial justice.

4.5 Public and environmental health

Several components of this scenario already exist in urban health digital twins and environmental monitoring systems that integrate air quality sensors, de-identified hospital records, and spatial data to identify risk patterns. AUDiTs extend this approach by linking these datasets within adaptive, ethics-aware agents. The Data Curator Agent, which is technically feasible, supports compliance with privacy standards while enabling fine-grained modeling of health risks. Future versions may incorporate autonomous policy testing to evaluate how interventions such as green canopy expansion, traffic calming, or industrial zoning changes influence exposure and vulnerability. Although empirical implementation of multi-agent systems for public health is still limited, these scenarios outline a research trajectory that moves from current analytical frameworks toward more holistic and learning-centered approaches to public health governance.

5 Challenges

Emerging human-centered digital twins, autonomous GIS prototypes, and urban systems-of-systems simulations demonstrate that the vision of *AUDiTs* is rapidly moving from conceptual theory toward practical realization. These early pilots showcase how embedded intelligence, adaptive governance, and multi-agent interaction can transform the way cities are modeled, managed, and co-evolved. Yet, to translate AUDiTs from experimental platforms into sustainable instruments of *New Urban*

Science, their technical, ethical, institutional, and ecological challenges must be addressed in an integrated and anticipatory manner. The transition from early prototypes to mature AUDiTs depends not only on technological innovation but on responsible orchestration across ethics, energy, and institutions. By embedding agentic intelligence within accountable and sustainable design, AUDiTs can become credible instruments for reimagining urban governance, enabling cities that learn, adapt, and evolve in partnership with their inhabitants.

5.1 Data privacy and ethics

AUDIiTs rely on large scale integration of heterogeneous urban data, including mobility traces, energy usage, health records, and social media activity. This breadth of information raises significant concerns about privacy, consent, and digital surveillance. Balancing data granularity for modeling precision with respect for privacy and personal autonomy is essential, especially when representing vulnerable or marginalized populations. Embedding an Ethics and Safety Agent within the AUDiT framework can provide an important governance safeguard by supporting dynamic anonymization, privacy-preserving computation, and adaptive risk calibration that reflects the sensitivity of specific data sources. Yet technical safeguards alone cannot guarantee ethical legitimacy. That legitimacy must emerge from inclusive co-design, where communities affected by data use participate in defining appropriate applications, acceptable levels of risk, harm thresholds, and procedures for remediation. Institutional mechanisms such as urban data review boards, community oversight panels, and transparent disclosure protocols can help ensure that AUDiTs remain accountable to the public interest and do not create new forms of algorithmic exclusion or data colonialism.

5.2 Algorithmic bias and explainability

Because AUDiTs depend on AI and machine learning models trained on historical and spatially uneven datasets, they face the risk of reproducing or intensifying structural biases. Preventing these feedback loops requires a sustained commitment to algorithmic transparency, fairness, and interpretability throughout the entire system lifecycle, beginning with data acquisition and extending through modeling, validation, and decision support. The Dialogue Mediator Agent has an especially important role by facilitating two-way communication between human decision makers and machine intelligence. It can surface underlying model assumptions, quantify uncertainties, and support counterfactual exploration that reveals how model outputs change under alternative conditions. Human oversight should be

central rather than peripheral, with ethicists, social scientists, and community representatives participating in model design, evaluation, and continuous monitoring. Analytical tools such as SHAP, LIME, and causal explainers should serve as standard mechanisms for interpreting model reasoning and exposing the basis of predictions. Through these practices, AUDiTs can function as transparent and accountable partners in urban governance rather than opaque algorithmic authorities.

5.3 Compute sustainability and scalability

AUDiTs' capacity for continuous sensing, learning, and simulation entails significant computational demands. Training large AI models on high-frequency urban data streams is energy-intensive, producing substantial carbon footprints that contradict sustainability goals. Future-proofing AUDiTs requires Green AI strategies to reconcile computational performance with environmental responsibility. These strategies include federated and edge learning to minimize data movement, energy-aware algorithms, adaptive model compression, and hardware optimization. Scalability must also ensure equitable access: smaller cities or regions with limited computational infrastructure should be able to deploy lightweight AUDiT variants, avoiding an "innovation divide" where only data-rich metropolitan areas benefit from digital twin intelligence.

5.4 Institutional and regulatory alignment

AUDiTs operate across jurisdictional boundaries and data regimes, exposing gaps in urban governance, accountability, and liability. Effective deployment requires institutional innovation and regulatory foresight. Priority areas include standards for model validation, algorithmic accountability, data stewardship, and citizen redress. Governance frameworks must evolve alongside emerging norms of AI autonomy, data sovereignty, and digital rights. To ensure legitimacy and resilience, AUDiTs should be embedded within multi-stakeholder governance ecosystems—for example, through digital ethics review boards or AI oversight councils that include policymakers, technologists, and civil society representatives. Sustained cross-agency collaboration will be crucial to ensure that AUDiTs remain transparent, interoperable, and democratically governed instruments—augmenting urban governance rather than existing as isolated technological artifacts.

6 Conclusion

While collaborative and participatory approaches are essential to democratic urban governance, they are often constrained by cognitive and temporal limits. Stakeholders can only process a small fraction of the information

flows that characterize complex urban systems. AUDiTs do not replace participatory processes. They augment them by synthesizing heterogeneous data, simulating multi scalar feedback, and revealing systemic interdependencies that remain beyond the reach of human deliberation. Recent advances in explainable urban AI, including the use of SHAP based interpretability to clarify the contributions of morphological and environmental determinants in complex urban systems, demonstrate how computational tools can strengthen analytical capacity in ways that manual methods cannot achieve (Eshraghi et al., 2025).

In this way, AI-assisted planning complements rather than substitutes collaboration: it expands the scope of participation from qualitative dialogue to quantitative, evidence-based co-learning, where diverse actors can test hypotheses, visualize trade-offs, and interrogate the assumptions underlying decisions. Unlike conventional participatory methods that rely on static representations or expert-driven facilitation, AUDiTs provide interactive, explainable, and traceable decision environments that support transparency, reproducibility, and context sensitive reasoning. This orientation aligns with emerging theories that frame cities as complex, open, and evolving social ecological technological systems, where resilience is understood as a continuous process of learning and adaptation rather than a fixed capacity (Esposito, 2025). Through embedded agents such as the Dialogue Mediator and the Ethics and Safety Agents, AUDiTs preserve the participatory ethos of planning by facilitating dialogue, accountability, and fairness, while using computational intelligence to strengthen safety, governance quality, and informed collective action.

To clarify scope, this paper positions AUDiTs as a forward-looking research framework rather than a fully realized system. Some components, such as autonomous GIS engines, GeoAI modeling, and participatory interfaces, are already emerging in prototype form. Other capabilities, including multiagent orchestration and ethical self-governance, remain aspirational and point toward future research frontiers. The intention is not to assert technological maturity, but to outline a research trajectory that connects current progress with longer-term societal and technical goals.

At the intersection of autonomous GIS, GeoAI, and agent based orchestration, AUDiTs point toward a new class of computational tools capable of scalable data fusion and context aware decision support. Some envisioned capabilities, such as LLM augmented GIS or harmonizing structured and unstructured data into unified geospatial representations, are not yet realized as established frameworks. Their absence highlights fertile ground for research and innovation. AUDiTs should

therefore be viewed as emerging prototypes and invitations to develop the connective tissue across methods, models, and data types. They offer a platform for integrating physics-based models with human centric and symbolic reasoning. This creates new opportunities to address spatially and temporally complex urban challenges, consistent with recent work that frames cities as adaptive and learning based social ecological technological systems (Esposito, 2025). By building on the predictive and planning functions that already exist in digital twins, AUDiTs extend their scope through EDUG principles, providing prediction, explanation, dialogue, uncertainty modeling, and goal sensitive simulation in ways that support iterative, anticipatory, and transformative forms of urban governance.

As cities confront escalating climate, mobility, health, and equity challenges, AUDiTs represent both an urgent opportunity and a significant research frontier. When embedded within living laboratories and governance systems, they can support participatory policy testing in simulated, data informed environments while strengthening ethical and regulatory preparedness. Realizing this vision requires careful attention to risks such as model misalignment, drift, hallucination, data ownership, and the misuse of realistic synthetic simulations, along with persistent concerns about privacy, bias, and fairness. Responsible AI frameworks, including guidance from the NIST AI Risk Management Framework, the OECD AI Principles, and the EU AI Act, can provide structure for testing, compliance, and emergency protocols. Through iterative human and AI co-learning among domain experts, developers, and urban stakeholders, AUDiTs can evolve from technical tools into responsible epistemic infrastructures that democratize advanced spatial reasoning and advance accountability, transparency, and public trust.

AUDI-Ts can be evaluated through measurable and transparent performance criteria. Early prototypes can be assessed using indicators such as the predictive accuracy of urban simulations, the stability of data fusion pipelines, the responsiveness of agentic orchestration under increasing scenario complexity, and levels of user trust and comprehension within participatory interfaces. Uncertainty reporting should include calibration plots, confidence intervals, and sensitivity analyses that show how assumptions propagate through interconnected modeling components. Fairness and safety reviews should examine potential disparate impacts across demographic or spatial groups, the clarity and traceability of decision pathways, and alignment with established Responsible AI frameworks. Together, these practices offer an initial foundation for systematic evaluation, enabling AUDiT prototypes to be tested,

audited, and refined within living laboratories and other real world settings.

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Authors' contributions

Xinyue Ye conceptualized the study and wrote the main manuscript text. Wenjing Gong co-developed the main manuscript text and prepared Fig. 1. All authors contributed to the writing and edited the manuscript.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

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